

--- SCORE SUMMARY ---

Selected Mark Scheme: Engineering

Overall Grade: 82/100

Grammar & Readability: 9/10

Plagiarism Probability: 10%

AI Usage Probability: 15%

--- DETAILED SECTION FEEDBACK ---

CONTEXT AIMS OBJECTIVES:

The project clearly defines a problem (safety in airspace) and establishes specific aims and objectives related to developing a classification system. The challenge level is appropriately framed, though the focus could be slightly more critical regarding the limitations of current methods. [Marks: 5/5]

LITERATURE REVIEW:

The review covers relevant deep learning and computer vision techniques for object detection, but could delve deeper into the specific challenges of aerial object classification (e.g., occlusions, scale variations, low altitude targets). The connection between existing work and the proposed solution is somewhat implicit. [Marks: 3/5]

METHODOLOGICAL APPROACH:

The approach section describes the use of CNNs and transformers effectively, but could better justify the chosen architectures (ResNet-50, MobileNetV3, ConvNeXt-Tiny) and explain why they are suitable for this specific problem. The lifecycle description is adequate but lacks detail on how different stages were managed. [Marks: 3/5]

REQUIREMENTS SPECIFICATION:

The objectives are well-defined (distinguish drones/birds/planes/miscellaneous objects). However, specific performance requirements (beyond general accuracy/latency) could be clearer. The justification for the system's functionality is sound. [Marks: 4/5]

IT DESIGN ANALYSIS:

The design choices (ConvNeXt-Tiny for accuracy, MobileNetV3 for speed) are well-justified. The system architecture and rationale are clearly presented. The dataset preparation process is described but could be more detailed regarding the specific challenges overcome. [Marks: 4/5]

IMPLEMENTATION DISCUSSION:

The implementation section effectively covers the comparison of different models (baseline vs. fine-tuned) and highlights the performance trade-offs (accuracy vs. latency). The discussion on engineering challenges (dataset preparation, preprocessing) is good but could elaborate more on the specific techniques used for augmentation and tuning. [Marks: 4/5]

VERIFICATION VALIDATION:

The project mentions testing and evaluation metrics, but details on the specific testing methodology (test sets, validation

strategy) are limited. The justification for the evaluation approach could be stronger. [Marks: 3/5]

EVALUATION AGAINST REQUIREMENTS:

The evaluation evidence (accuracy >96%, speed) strongly supports meeting the core requirements. However, limitations (e.g., dataset bias, generalization to unseen conditions) are acknowledged but not deeply analyzed. [Marks: 4/5]

PROJECT PLANNING MANAGEMENT:

No explicit project plan or management details are provided in the abstract/summary. This section is missing entirely. [Marks: 0/5]

SOLUTION ATTRIBUTES:

The system demonstrates high accuracy and speed, showcasing good reliability and performance. The dataset preparation challenges highlight an area for improvement regarding completeness/maintainability. Overall, the artefact is technically sound. [Marks: 5/5]

SUMMARY CONCLUSIONS RECOMMENDATIONS:

The conclusions effectively summarize the findings (ConvNeXt-Tiny's accuracy, MobileNetV3's speed). Recommendations for future work (e.g., addressing dataset limitations, improving robustness) are present but could be more specific. [Marks: 4/5]

STRUCTURE PRESENTATION:

The report structure is logical, and the writing is clear and concise. Diagrams (if included) would enhance readability, but the current presentation is acceptable. [Marks: 4/5]

OVERALL UNDERSTANDING REFLECTION:

The project demonstrates a good understanding of deep learning and computer vision concepts. There is some reflection on challenges (dataset, model limitations), but deeper critical insight regarding the broader implications (e.g., ethical considerations, societal impact) is lacking. [Marks: 3/5]